MATH 152
Mrs. Bonny Tighe

# EXAM III

PART ONE
11.8-10.1
50 points

NAME Solutions

TA_____ Wed. 11/28/2018

1. (4 points each) Circle the letter for the correct answer

a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n+1}}{9^n (2n+1)!}$

$$\sum_{n=0}^{\infty} (-1)^n \frac{\pi^{2n+1} (3)}{3^{2n} \cdot 3 \, (2n+1)!}$$

$$3 \sum_{n=0}^{\infty} (-1)^n \frac{(\pi/3)^{2n+1}}{(2n+1)!}$$

$$3 \sin \pi/3$$

A) $3/2$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/6$

D) $\sin(1/2)$

E) Some other answer

b) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n^2+2}$

$$\left| (x-1) \frac{(n+1)^2+2}{n^2+2} \right| < 1$$

$$|x-1| < 1$$

$$-1 < x-1 < 1$$

$$0 < x < 2$$

A) $(0,2)$

B) $(0,2]$

C) $[0,2]$

D) $[0,2)$

E) Some other answer

c) The Maclaurin series representation for $f(x) = \tan^{-1}(3x^2)$ is:

$$\sum_{n=0}^{\infty} (-1)^n \frac{(3x^2)^{2n+1}}{2n+1}$$

$$\sum_{n=0}^{\infty} (-1)^n \frac{3^{2n+1} x^{4n+2}}{2n+1}$$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n 3x^{4n+2}}{2n}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{2n+1}}{2n++1}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+2}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+2}}{2n+1}$

E) Some other answer

d) Which of the following parabola has its vertex at (2, 0) and the directrix of $x = 1$

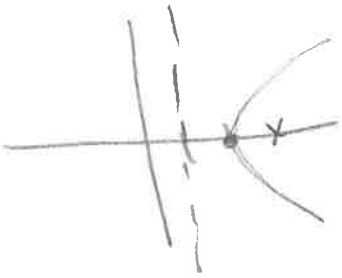
A) $(x-2)^2 = 4y$

B) $(x-2)^2 = -4y$

C) $y^2 = 4(x-2)$

D) $y^2 = -4(x-2)$

E) Some other answer



e) The series which is equivalent to $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is:

$$\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{(2n+2)!}$$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{(2n+1)!}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{(2n+2)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n}}{(2n-1)!}$

E) Some other answer

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$

E) Some other answer

g) $T_6(x)$ for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{4^{n+1}}$ is

$\frac{1}{4} - \frac{x^2}{4^2} + \frac{x^4}{4^3} - \frac{x^6}{4^4}$

A) $\frac{1}{4} - \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 - \frac{1}{4^4}x^6 + \frac{1}{4^5}x^8 - \frac{1}{4^6}x^{10}$

B) $\frac{1}{4} - \frac{1}{4^2} + \frac{1}{4^3} - \frac{1}{4^4} + \frac{1}{4^5} - \frac{1}{4^6}$

C) $\frac{1}{4} + \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 + \frac{1}{4^4}x^6$

D) $\frac{1}{4} - \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 - \frac{1}{4^4}x^6$

E) Some other answer

2. (2 points each) True or False

a) TRUE FALSE $4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 = \frac{8!}{3!}$

b) TRUE FALSE The Maclaurin series for $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n}}{(2n)!}$

c) TRUE FALSE The asymptotes for the hyperbola $\frac{(y-1)^2}{4} - \frac{(x+2)^2}{9} = 1$ are $y = \pm \frac{2}{3}x$

d) TRUE FALSE Given
$f(2) = 1$, $f'(2) = -2/3$, $f''(2) = 2 \cdot 5/3^2$, $f'''(2) = -2 \cdot 5 \cdot 8/3^3$, etc.

then $f^{(n)}(2) = \frac{(-1)^n (2 \cdot 5 \cdot 8 \cdots (3n-1)}{3^n}$

e) TRUE FALSE If $-2 < x < 4$ then $|x-2| < 2$

$-4 < x-2 < 2$

f) TRUE FALSE $\frac{1}{x-1} = \sum_{n=1}^{\infty} \frac{1}{x}\left(\frac{1}{x}\right)^n$

g) TRUE FALSE The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^n}{3^{2n}}$ is 3.

h) TRUE FALSE $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = \frac{1}{e}$

i) TRUE FALSE The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ is 1.

3. (9 points) Use the Binomial Theorem to find the function as a power series.

$$f(x) = \frac{1}{\sqrt{1-x^3}} = \left(1 + (-x^3)\right)^{-1/2} \qquad \sum_{n=0}^{\infty} \square \frac{(-x^3)^n}{n!}$$

$$\square = 1 - \tfrac{1}{2} - \tfrac{1}{2}\left(-\tfrac{3}{2}\right) - \tfrac{1}{2}\left(-\tfrac{3}{2}\right)\left(-\tfrac{5}{2}\right)$$

$$= \underset{n=0}{1} - \underset{n=1}{\frac{1}{2}} + \underset{n=2}{\frac{1\cdot 3}{2^2}} - \underset{n=3}{\frac{1\cdot 3\cdot 5}{2^3}} = \frac{(-1)^n(1\cdot 3\cdot 5\cdots(2n-1))}{2^n} \leftarrow n=0 \text{ outside } \Sigma$$

$$1 + \sum_{n=1}^{\infty} \frac{\cancel{(-1)^n}(1\cdot 3\cdot 5\cdots(2n-1))}{2^n} \frac{\cancel{(-1)^n}\, x^{3n}}{n!}$$

$$1 + \sum_{n=1}^{\infty} \frac{(1\cdot 3\cdot 5\cdots(2n-1))}{2^n\, n!} x^{3n}$$

MATH 152 **EXAM III** NAME Solutions

Mrs. Bonny Tighe PART TWO

11.8-10.1 TA_____ Wed.11/28/2018

50 points

1. (4 points) Find the vertex, focus, and directrix. Sketch the graph

$y^2 + 4y + 2x + 2 = 0$ $+4$

$(y+2)^2 = -2x - 2 + 4$

$(y+2)^2 = -2x + 2$

$(y+2)^2 = -2(x-1)$

focal width $= 2$

$p = 1/2$

vertex $(1,-2)$

focus $(1/2, -2)$

directrix $x = 3/2$

2. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$9(x+2)^2 - 4(y-1)^2 = 36$

$\frac{(x+2)^2}{4} - \frac{(y-1)^2}{9} = 1$

center $(-2, 1)$

foci $(-2 \pm \sqrt{13}, 1)$

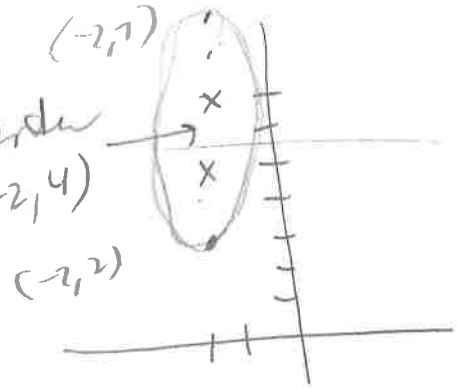
vertices $(-2 \pm 2, 1)$

asymptotes $(y-1) = \pm \frac{3}{2}(x+2)$

3. (4 points) Find the equation of the ellipse with vertex at (-2, 3) and foci at (-2, 5) and (-2, 7)

$\frac{(x+2)^2}{8} + \frac{(y-4)^2}{9} = 1$

center $(-2, 4)$




$c = 1$

$c^2 = a^2 - b^2$

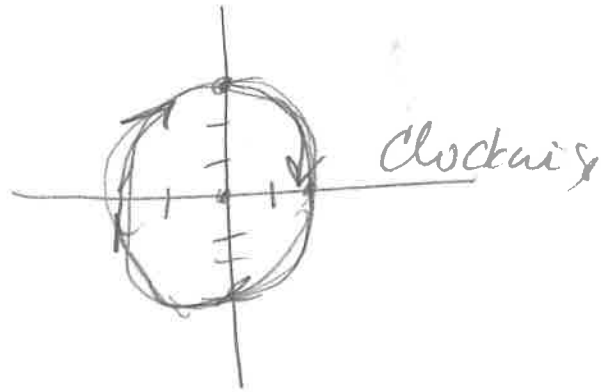
$1 = 9 - b^2$

$b^2 = 8$

4. (5 points) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 2\sin\theta, \quad y = 3\cos\theta$

$$\frac{x^2}{4} + \frac{y^2}{9} = 1$$

| $\theta$ | x | y |
|---|---|---|
| 0 | 0 | 3 |
| $\pi/2$ | 2 | 0 |




5. (5 points) Use series to evaluate the limit $\lim_{x\to 0} \dfrac{1-\cos(x^2)}{e^{x^2}-1-x^2}$

$$\cos(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n}}{(2n)!} = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!}$$

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = 1 + \frac{x^2}{1} + \frac{x^4}{2!} + \frac{x^6}{4!}$$

$$\lim_{x\to 0} \left( \frac{1-\left(1-\frac{x^4}{2}+\frac{x^8}{4!}-\frac{x^{12}}{6!}\cdots\right)}{\left(1+x^2+\frac{x^4}{2}+\frac{x^6}{4}\right)-1-x^2} \right.$$

$$\lim_{x\to 0} \frac{\frac{x^4}{2} - \frac{x^8}{4!} + \frac{x^{12}}{6!}\cdots}{\frac{x^4}{2} + \frac{x^6}{4}\cdots} = 1$$

86 (10 points) Find the Taylor series for $f(x) = \dfrac{1}{(3+x)^3}$ centered at

a = 2

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(2)\,(x-2)^n}{n!}$$

$f(x) = (3+x)^{-3}$

$f'(x) = -3(3+x)^{-4}$

$f''(x) = 3\cdot 4(3+x)^{-5}$

$f'''(x) = -5\cdot 4\cdot 3(3+x)^{-6}$

$f(2) = \frac{1}{5^3}$

$f'(2) = \frac{-3}{5^4}$

$f''(2) = \frac{3\cdot 4}{5^5}$

$f'''(2) = \frac{-5\cdot 4\cdot 3}{5^6}$

$\frac{1}{5^3} - \frac{3!}{2\cdot 5^4} + \frac{4!}{2\cdot 5^5} - \frac{5!}{2\cdot 5^6}$

$\frac{(-1)^n (n+2)!}{2\cdot 5^{n+3}}$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (n+2)(n+1)}{2\cdot 5^{n+3}} (x-2)^n$$

7. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{3^n (n+2)}$$

$R = 3$

$I = (-1, 5]$

$$\lim_{n\to\infty} \left| \frac{(x-2)\,(x-2)^n}{3\cdot 3^n (n+3)} \cdot \frac{3^n (n+2)}{(x-2)^n} \right|$$

$$\lim_{n\to\infty} \left| \frac{(x-2)}{3} \cdot \frac{n+2}{n+3} \right| = \left| \frac{x-2}{3} \right| < 1$$

$|x-2| < 3$

$-3 < x-2 < 3$

$-1 < x-2 < 5$

test end points

$x = -1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (-3)^n}{3^n (n+2)} = \frac{1}{n+2} \text{ diverges}$$

$x = 5$

$$\sum_{n=0}^{\infty} \frac{(-1)^n (3^n)}{3^n (n+2)} \quad \frac{(-1)^n}{n+2} \text{ Cond. Converg.}$$

8. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x)=\sqrt{x}$ centered at $a=9$.

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$

$f(x) = x^{1/2}$ $\quad f(9) = 3$

$f'(x) = \frac{1}{2}x^{-1/2}$ $\quad f'(9) = \frac{1}{2}\cdot\frac{1}{3}$

$f''(x) = -\frac{1}{2^2}\cdot x^{-3/2}$ $\quad f''(9) = -\frac{1}{2^2}\cdot\frac{1}{3^3}$

$f'''(x) = \frac{1\cdot 3}{2^3}x^{-5/2}$ $\quad f'''(9) = \frac{1\cdot 3}{2^3}\cdot\frac{1}{3^5}$

$$T_3(x) = 3 + \frac{1}{6}(x-9) - \frac{1}{2^2\cdot 3^3}\frac{(x-9)^2}{2!} + \frac{3}{2^3\cdot 3^5}\frac{(x-9)^3}{3!}$$

b) (4 points) Use it to approximate $\sqrt{9.1}$.

$$\sqrt{9.1} = 3 + \frac{1}{6}(.1) - \frac{1}{2^2 3^3}\frac{(.1)^2}{2!} + \frac{3}{2^3\cdot 3^5}\frac{(.1)^3}{3!}$$

$8.5 < x < 9.5$

$-.5 < x-9 < .5$

c) (4 points) How accurate is this approximation when $|x-9|<0.5$

If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

$|f''''(x)| = \left|\frac{15}{2^4}x^{-7/2}\right|$

$M \ge \left|\frac{15}{16(8.5)^{7/2}}\right|$

$\frac{15}{16x^{7/2}}$

largest at $x = 8.5$

$$R_4 \le \frac{\left|\frac{15}{16(8.5)^{7/2}}\right|(.5)^4}{4!}$$

9. (5 points) Evaluate the integral as a power series and approximate to $T_6$.

$$\int_0^{0.5} \frac{x}{4-x^2}\,dx$$

$$\int_0^{0.5} \frac{\frac{x}{4}}{1-\frac{x^2}{4}}\,dx = \int_0^{0.5} \sum_{n=0}^{\infty} \frac{x}{4}\left(\frac{x^2}{4}\right)^n dx$$

$$\int_0^{0.5} \sum_{n=0}^{\infty} \frac{x^{2n+1}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{x^{2n+2}}{4^{n+1}(2n+2)}\Bigg]_0^{0.5}$$

$$\frac{x^2}{4(2)} + \frac{x^4}{4^2(4)} + \frac{x^6}{4^3(6)}\Bigg]_0^{0.5}$$

$$\left(\frac{(0.5)^2}{8} + \frac{(0.5)^4}{4^3} + \frac{(0.5)^6}{4^3(6)}\right) - (0)$$

MATH 152 **EXAM IIIA** NAME Solutions

Mrs. Bonny Tighe PART ONE

11.8-10.2 TA_____Wed.10/28//2018

50 points

1. (4 points each) Circle the letter for the correct answer

a) The Maclaurin series representation for $f(x) = x\cos(x^3)$ is:

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n+1)!}$

B) $\sum_{n=0}^{\infty} \frac{x^{6n+1}}{2n+1}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{2n+1}$

E) Some other answer

$\sum_{n=0}^{\infty} x \frac{(-1)^n (x^3)^{2n}}{(2n)!}$

$\sum_{n=0}^{\infty} \frac{(-1)^n (x^{6n+1})}{(2n)!}$

b) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n+1}}{3^{2n}(2n+1)!}$

$\frac{1}{3^{2n+1}}(3) = 3\sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n+1}}{(2n+1)!}$

A) $\pi/3$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/2$

D) $1/2$

E) Some other answer

$3 \sin \pi/3$

$3(\sqrt{3}/2)$

c) The radius of convergence for the series $\sum_{n=1}^{\infty} \frac{3^n x^n}{7^{n+1}}$

A) $3/7$

B) $(-\infty, \infty)$

C) $7/3$

D) $\{0\}$

E) Some other answer

$\lim_{x \to \infty} \left| \frac{3 \cdot 3^n x \cdot x^n}{7 \cdot 7^{n+1}} \cdot \frac{7^{n+1}}{3^n \cdot x^n} \right| =$

$\left| \frac{3}{7} x \right| < 1$

$|x| < 7/3$

d) Which of the following parabola has its vertex at (0, 0) and the directrix of $y = -1$

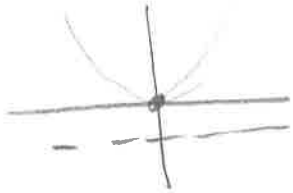
A) $x^2 = 4y$

B) $x^2 = -4y$

C) $x^2 = -4y$

D) $y^2 = -4x$

E) Some other answer



$x^2 = 4py$

e) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n)!}$

C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

D) $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$

E) Some other answer

f) The series which is equivalent to $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{3^n n!}$

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{3^n n!}$

$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{3^{n+1}(n+1)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{3^{n+1} (n+1)!}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n-1}}{3^{n-1} (n-1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{3^{n+1} (n+1)!}$

E) Some other answer

g) $T_5(x)$ for the series $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n+1}}{3^n\, n!}$ is

$\frac{x}{3} - \frac{x^3}{3(1)} + \frac{x^5}{9(2)}$

A) $x - \frac{x^3}{3} + \frac{x^5}{18} - \frac{x^7}{27(6)} + \frac{x^9}{81(24)}$

B) $x - \frac{x^3}{3} + \frac{x^5}{18} - \frac{x^7}{27(6)}$

C) $x - \frac{x^3}{3} + \frac{x^5}{9}$

D) $x - \frac{x^3}{3} + \frac{x^5}{18} - \cancel{\frac{x^7}{27}} + \cancel{\frac{x^9}{81}}$

E) Some other answer

2. (2 points each) True or False Circle the correct answer

a) TRUE FALSE $\frac{1}{x-1} = \sum_{n=0}^{\infty} \left(\frac{1}{x}\right)\left(\frac{1}{x}\right)^n$

b) TRUE FALSE The Maclaurin series for $\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n+1}$

c) TRUE FALSE The asymptotes for the hyperbola $\frac{(x-1)^2}{4} - \frac{(y+2)^2}{9} = 1$ are $y = \pm \frac{3}{2} x$

d) TRUE FALSE Given $f(2) = 1$, $f'(2) = 2/3$, $f''(2) = -2\cdot 5/3^2$, $f'''(2) = 2\cdot 5\cdot 8/3^3$, etc. then $f^{(n)}(2) = \frac{(-1)^n (2\cdot 5\cdot 8 \cdots (3n-1)}{3^n}$

e) TRUE FALSE If $-3 < x < 5$ then $|x+1| < 2$

$-2 < x+1 < 6$

f) TRUE FALSE The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^n\, n!}$ is 3.

g) TRUE FALSE $\frac{5!}{2} = 5\cdot 4\cdot 3\cdot 2\cdot 1$

h) TRUE FALSE $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}$

i) TRUE FALSE The Taylor series for $f(x) = \sin x$ at $x = \pi/3$ is $\sum_{n=0}^{\infty} \frac{(-1)^n (x - \pi/3)^{2n+1}}{(2n+1)!}$.

3. (9 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \frac{1}{(1-x^3)^3}$ $= (1-x^3)^{-3} = (1+(-x^3))^{-3}$ $\sum_{n=0}^{\infty} \square \frac{(-x^3)^n}{n!}$

$\square = 1 - 3 - 3(-4) - 3(-4)(-5)$

$= 1 - 3 + 3\cdot 4 - 3\cdot 4\cdot 5 + 3\cdot 4\cdot 5\cdot 6$

$(-1)^n \frac{(n+2)!}{2}$ $\sum_{n=0}^{\infty} (-1)^n \frac{(n+2)!}{2} \frac{(-1)^n x^{3n}}{n!}$

$\sum_{n=0}^{\infty} \frac{\cancel{(-1)^n}(n+2)(n+1)\cancel{n!}\cancel{(-1)^n}}{2} \frac{x^{3n}}{\cancel{n!}}$

$$\boxed{\sum_{n=0}^{\infty} \frac{(n+2)(n+1)}{2} x^{3n}}$$

MATH 152 **EXAM IIIA** NAME Solutions

Mrs. Bonny Tighe PART TWO

11.8-10.1 TA_____ Wed.5/2/2018

50 points

1. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \frac{1}{\sqrt{x}}$ centered at $a = 4$.

$f(x) = x^{-1/2}$ | $f(4) = \frac{1}{2}$

$f'(x) = -\frac{1}{2}x^{-3/2}$ | $f'(4) = -\frac{1}{2} \cdot \frac{1}{2^3}$

$f''(x) = \frac{3}{4}x^{-7/2}$ | $f''(4) = \frac{3}{4} \cdot \frac{1}{2^5}$

$f'''(x) = \frac{-15}{8}x^{-9/2}$ | $f'''(4) = \frac{-15}{8} \cdot \frac{1}{2^7}$

$$T_3(x) = \frac{1}{2} - \frac{1}{16}(x-4) + \frac{3}{2^7}\frac{(x-4)^2}{2!} - \frac{15}{2^{10}}\frac{(x-4)^3}{3!}$$

b) (4 points) Use it to approximate $\frac{1}{\sqrt{3.9}}$.

$$\frac{1}{\sqrt{3.9}} \approx \frac{1}{2} - \frac{1}{16}(3.9-4) + \frac{3}{2^7}\frac{(3.9-4)^2}{2} - \frac{15}{2^{10}}\frac{(3.9-4)^3}{3!} \quad \text{or}$$

$$= \frac{1}{2} - \frac{1}{16}\left(-\frac{1}{10}\right) + \frac{3}{2^7}\left(-\frac{1}{10}\right)^2 \cdot \frac{1}{2} - \frac{1}{2^{10}}\left(-\frac{1}{10}\right)^3 \frac{1}{3!}$$

c) (4 points) How accurate is this approximation when $|x-4| < 0.5$

$-.5 < x-4 < .5$

$3.5 < x < 4.5$

If $|f^{(n+1)}(x)| \le M$ for $|x-a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x-a|^{n+1}$ for $|x-a| \le d$

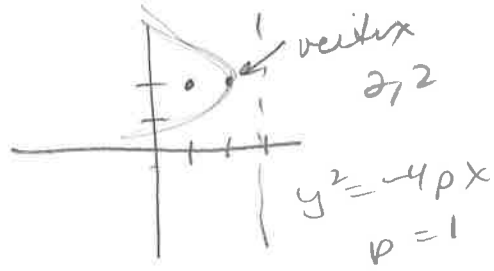
$f''''(x) = \left|\frac{105}{16}x^{-11/2}\right|$ largest when $x = 3.5$

$M \ge \left|\frac{105}{16(3.5)^{11/2}}\right|$

$$R_4(x) \le \frac{105}{16(3.5)^{11/2}}\left(\frac{|.5|^4}{4!}\right)$$

2. (4 points) Find the equation of the parabola with directrix of $x = 3$ and focus at $(1,2)$.

$(y-2)^2 = -4(x-2)$




3. (4 points) Find the center, foci, and vertices. Sketch the graph

$4x^2 + 16x + 9y^2 + 36y + 16 = 0$

$4(x^2+4x+4) + 9(y^2+4y+4) = -16+16+36$

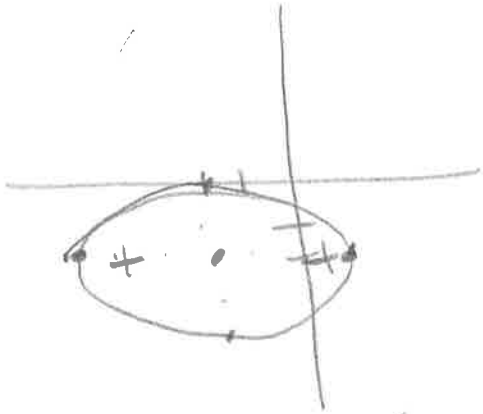
$4(x+2)^2 + 9(y+2)^2 = 36$

$\frac{(x+2)^2}{9} + \frac{(y+2)^2}{4} = 1$

center $(-2,-2)$

foci $(-2 \pm \sqrt{5}, -2)$

vertices $(-2 \pm 3, -2)$



4. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.

$16(y-2)^2 - 25(x+1)^2 = 400$

$\frac{(y-2)^2}{25} - \frac{(x+1)^2}{16} = 1$

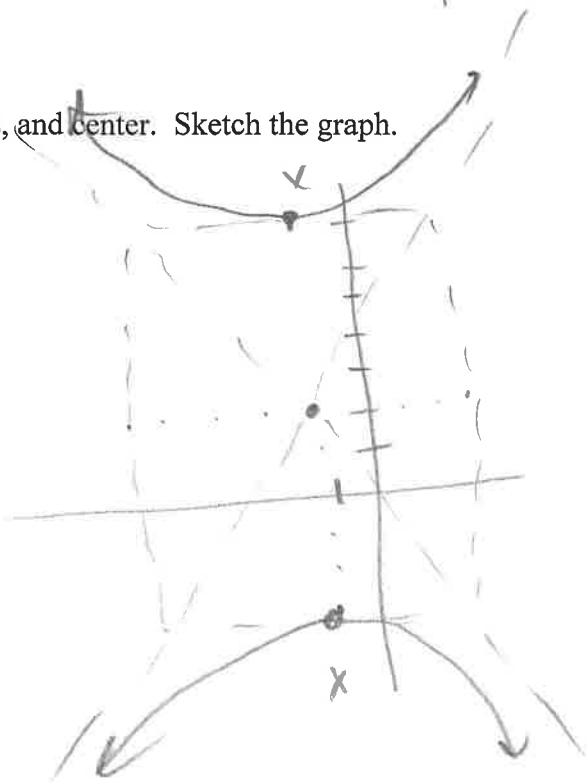
center $(-1, 2)$

vertices $(-1, 2 \pm 5)$

foci $(-1, 2 \pm \sqrt{41})$

asymptotes

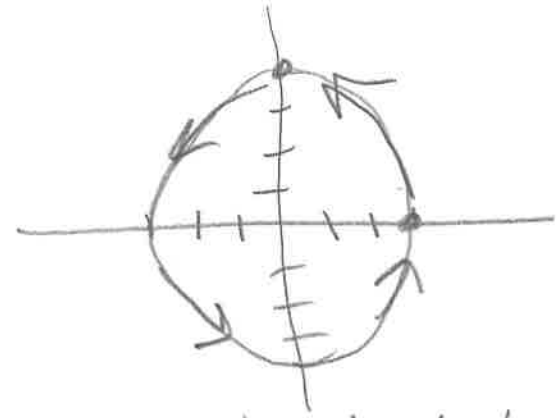
$(y-2) = \pm \frac{5}{4}(x+1)$



5. (5 points) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 3\cos\theta, \quad y = 4\sin\theta$

$$\frac{y^2}{16} + \frac{x^2}{9} = 1$$

| $\theta$ | $x$ | $y$ |
|---|---|---|
| 0 | 3 | 0 |
| $\pi/2$ | 0 | 4 |



Counterclockwise

6. (5 points) Use series to evaluate the limit $\lim_{x\to 0} \frac{1-\cos(x^2)}{e^{x^2}-1-x^2}$

$$\cos(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n}}{(2n)!} = 1 - \frac{x^4}{1} + \frac{x^8}{2!} - \frac{x^{12}}{4!} \cdots$$

$$e^{(x^2)} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = 1 + \frac{x^2}{1} + \frac{x^4}{2!} + \frac{x^6}{3!} \cdots$$

$$\lim_{x\to 0} \frac{1 - \left(1 - x^4 + \frac{x^8}{2} - \frac{x^{12}}{4!} \cdots\right)}{\left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} \cdots\right) - 1 - x^2} =$$

$$\lim_{x\to\infty} \left( \frac{x^4 - \frac{x^8}{2} + \frac{x^{12}}{4!} \cdots}{\frac{x^4}{2} + \frac{x^6}{3!} + \cdots} \right) =$$

$$\lim_{x\to\infty} \left( \frac{1 - \frac{x^4}{2} + \frac{x^8}{4!} \cdots}{\frac{1}{2} + \frac{x^2}{3!} + \cdots} \right) = \frac{1}{1/2} = 2$$

7. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x+2)^n}{\sqrt{n+2}}$$

$R=1$

$I=(-3,-1]$

$$\lim_{n\to\infty} \left| \frac{(x+2)\cancel{(x+2)^n}}{\sqrt{n+2}} \cdot \frac{\sqrt{n+2}}{\cancel{(x+2)^n}} \right| = \lim_{n\to\infty} \left| (x+2)\frac{\sqrt{n+2}}{\sqrt{n+2}} \right| = |x+2| < 1$$

$-1 < x+2 < +1$

$-3 < x < -1$

at $x=-3$

$$\sum \frac{(-1)^n(-1)^n}{\sqrt{n+2}}$$

$$\sum \frac{1}{\sqrt{n+2}}$$ Diverges

$x=-1$

$$\sum_{n=0}^{\infty} \frac{(-1)^n 1^n}{\sqrt{n+2}}$$

$$\sum \frac{(-1)^n}{\sqrt{n+2}}$$ Cond. converges

8. (10 points) Find the Taylor series for $f(x) = \dfrac{1}{(x+1)^2}$ centered at

a = 3

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(3)}{n!}(x-3)^n$$

| | |
|---|---|
| $f(x) = (x+1)^{-2}$ | $f(3) = \frac{1}{4^2}$ |
| $f'(x) = -2(x+1)^{-3}$ | $f'(3) = \frac{-2}{4^3}$ |
| $f''(x) = 3\cdot 2(x+1)^{-4}$ | $f''(3) = \frac{3!}{4^4}$ |
| $f'''(x) = -4!(x+1)^{-5}$ | $f'''(3) = \frac{-4!}{4^5}$ |

$$\underset{n=0}{\frac{1}{4^2}} - \underset{n=1}{\frac{2!}{4^3}} + \underset{n=2}{\frac{3!}{4^4}} - \underset{n=3}{\frac{4!}{4^5}}$$

$$\frac{(-1)^n (n+1)!}{4^{n+2}} \cdot \frac{(x-3)^n}{n!}$$

$$\boxed{\sum_{n=0}^{\infty} \frac{(-1)^n (n+1)(x-3)^n}{4^{n+2}}}$$

9. (5 points) Evaluate the integral as a power series and approximate to $T_6$.

$$\int_0^{0.5} \frac{x}{4+x^2}\,dx$$

$$\int_0^{0.5} \frac{\frac{x}{4}}{1-\left(-\frac{x^2}{4}\right)}\,dx \qquad \int_0^{0.5} \frac{x}{4} \sum_{n=0}^{\infty} \left(-\frac{x^2}{4}\right)^n dx$$

$$\int_0^{0.5} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}\,dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+2}}{4^{n+1}(2n+2)} \Bigg]_0^{0.5}$$

$$\frac{x^2}{4(2)} - \frac{x^4}{4^2(4)} + \frac{x^6}{4^3(6)} \Bigg]_0^{0.5}$$

$$\frac{(0.5)^2}{8} - \frac{(0.5)^4}{4^3} + \frac{(0.5)^6}{4^3(6)} = 0$$